

DEPARTMENT OF ELECTRICAL ENGINEERING

Electronic Devices & Circuits Laboratory (EDC LAB)

EXPERIMENT 5: MOSFET Differential Amplifier — DC Bias, Common-Mode and Differential-Mode Transfer Characteristics

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1. Aim and Objectives

1. To assemble and verify the operation of a two-transistor source-coupled MOSFET differential amplifier utilizing a matched SI9926BDY dual N-channel MOSFET package.
2. To characterize the individual input (I_D vs. V_{GS}) and output (I_D vs. V_{DS}) DC characteristics of both transistors (M_1 and M_2) inside the dual package.
3. To establish, verify, and document the DC operating (quiescent) point by adjusting the common-mode gate bias such that a nominal tail bias current $I_{SS} \approx 80$ mA is achieved with $V_{DD} = 12$ V, $R_D = 180\ \Omega$, and $R_{SS} = 22\ \Omega$.
Note: These values of resistances were chosen based on availability in lab.
4. To experimentally determine the common-mode transfer characteristics (V_{D1} , V_{D2} , and I_{SS} vs. V_{cm}) under pure common-mode excitation ($V_{id} = 0$), identifying the Input Common-Mode Range (ICMR) bounded by triode entry and cutoff.
5. To experimentally measure the large-signal differential-mode transfer characteristics (V_{D1} , V_{D2} , I_{D1} , I_{D2} , and $\Delta I_D = I_{D1} - I_{D2}$ vs. $V_{id} = V_{in+} - V_{cm}$) over a symmetric differential input span.
6. To test the exponential weak-inversion (subthreshold) conduction model on a semi-log scale ($\log |\Delta I_D|$ vs. V_{id}), extract the device slope factor n , and compare the measured small-signal transconductance g_m and voltage gain A_d against theoretical predictions.
7. To compare LTspice circuit simulations side-by-side with experimental waveforms, tabulating all measurements and performing comprehensive mismatch and CMRR calculations.

2. Apparatus and Circuit Specifications

Table 1: Equipment and Components Used in Experiment

S.No.	Component / Equipment	Value / Specification	Purpose / Operating Note
1	Dual N-Channel MOSFET	SI9926BDY (SO-8 to DIP-8)	Matched pair M_1, M_2 ($V_{DS,max} = 20$ V, $g_{fs} = 29$ S)
2	Drain Resistors (R_{D1}, R_{D2})	180 Ω , High Power	Symmetric drain loads ($V_{DD} = 12$ V)
3	Tail Resistor (R_{SS})	22 Ω , 1/4 W or higher	Source degeneration establishing tail current I_{SS}
4	Potentiometers	10 k Ω multi-turn	Input bias divider for V_{in+} and V_{in-}
5	Regulated DC Power Supply	0 – 30 V dual output	Set to $V_{DD} = 12$ V, current-limited at 150 mA
6	Digital Multimeters (DMM)	Precision 4.5 digit	Non-invasive voltage probes across R_{SS}, R_{D1}, R_{D2}
7	Simulation Software	LTspice XVII	Nonlinear SPICE simulation using SI9926 subcircuit

2.1 Pin Configuration and Schematic Diagram

The SI9926BDY dual TrenchFET resides in an SO-8 package mounted on a DIP-8 breakout board. Its pin connections are:

- **Pin 1:** Source 1 (S_1) — **Pin 2:** Gate 1 (G_1) — **Pin 3:** Source 2 (S_2) — **Pin 4:** Gate 2 (G_2)
- **Pins 5, 6:** Drain 2 (D_2) — **Pins 7, 8:** Drain 1 (D_1)

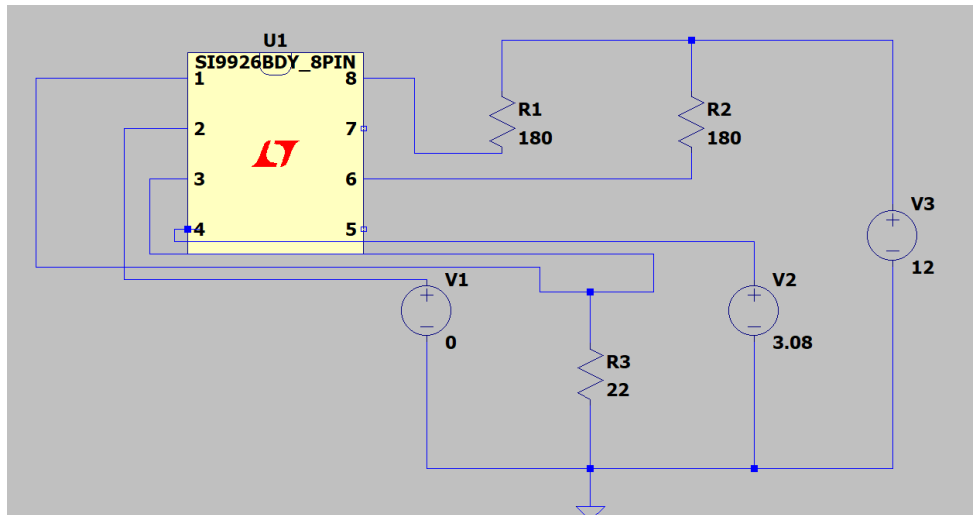


Figure 1: Circuit Schematic of the Source-Coupled MOSFET Differential Amplifier ($V_{DD} = 12$ V, $R_D = 180$ Ω , $R_{SS} = 22$ Ω).

3. Theoretical Framework

3.1 Differential Pair Conduction Physics: Strong vs. Weak Inversion

In a classic MOSFET differential pair, two symmetric transistors M_1 and M_2 have their sources tied together at node V_S and biased by a tail resistance R_{SS} . Drain currents are loaded symmetrically by

$R_{D1} = R_{D2} = R_D$. Kirchhoff's Current Law at the tail node mandates:

$$I_{D1} + I_{D2} = I_{SS} = \frac{V_S}{R_{SS}} \quad (1)$$

For an ideal square-law MOSFET in strong inversion, the large-signal differential current is governed by:

$$\Delta I_D = I_{D1} - I_{D2} = \beta V_{id} \sqrt{\frac{I_{SS}}{\beta} - \frac{V_{id}^2}{4}}, \quad \text{for } |V_{id}| \leq \sqrt{\frac{2I_{SS}}{\beta}} \quad (2)$$

where $\beta = \mu_n C_{ox}(W/L)$ is the transconductance parameter. However, the SI9926BDY is a vertical TrenchFET power device exhibiting an extraordinarily large transconductance parameter ($\beta \approx 51 \text{ A/V}^2$ derived from its rated $g_{fs} = 29 \text{ S}$). Achieving the classical strong-inversion criterion ($V_{OV} = V_{GS} - V_{th} \geq 100 \text{ mV}$) would necessitate tail currents on the order of multiple amperes ($I_{SS} = \frac{1}{2}\beta V_{OV}^2 \approx 250 \text{ mA}$ to 1 A).

At our lab bench quiescent bias of $I_{SS} \approx 80 \text{ mA}$ ($I_{D1} = I_{D2} \approx 40 \text{ mA}$), the nominal square-law overdrive is merely:

$$V_{OV0} = \sqrt{\frac{I_{SS}}{\beta}} = \sqrt{\frac{0.080}{51}} \approx 39.6 \text{ mV} \quad (3)$$

Because $V_{OV0} \ll 100 \text{ mV}$, the device operates predominantly in the **weak-inversion (subthreshold) / moderate-inversion regime**. In weak inversion, channel conduction is dominated by carrier diffusion rather than drift, yielding an exponential current-voltage relation analogous to a BJT:

$$I_D = I_0 \exp\left(\frac{V_{GS}}{nV_T}\right) \quad (4)$$

where $V_T = \frac{k_B T}{q} \approx 25.85 \text{ mV}$ at room temperature ($T = 300 \text{ K}$) and n is the subthreshold slope factor ($1.2 \leq n \leq 2.5$). Under this exponential formulation, the differential transfer characteristic becomes hyperbolic tangent:

$$\Delta I_D = I_{SS} \tanh\left(\frac{V_{id}}{2nV_T}\right) \quad (5)$$

The differential small-signal transconductance (differential transconductance of the pair) and differential voltage gain are:

$$g_m = \frac{I_{SS}}{2nV_T}, \quad A_d = g_m R_D \quad (6)$$

3.2 Common-Mode Gain and CMRR Formulation

With a finite resistive tail R_{SS} , applying a common-mode increment v_{cm} induces an incremental tail voltage v_s . Since the two transistors are in parallel for common-mode signals, the incremental common-mode gain at each drain terminal is:

$$A_{cm} = \frac{v_{d,cm}}{v_{in,cm}} \approx -\frac{g_m R_D}{1 + 2g_m R_{SS}} \approx -\frac{R_D}{2R_{SS}} \quad (\text{when } 2g_m R_{SS} \gg 1) \quad (7)$$

With $R_D = 180 \Omega$ and $R_{SS} = 22 \Omega$:

$$A_{cm} \approx -\frac{180}{2 \times 22} = -\frac{180}{44} \approx -4.091 \text{ V/V} \quad (8)$$

The Common-Mode Rejection Ratio (CMRR) is given by:

$$\text{CMRR} = \left| \frac{A_d}{A_{cm}} \right| \approx g_m R_{SS} \quad (9)$$

4. Device Characteristics of Matched Dual MOSFETs

Prior to assembling the differential amplifier, the input and output static characteristics of both transistors in the dual SI9926 package (designated Top Transistor V_1/M_1 and Bottom Transistor V_2/M_2) were experimentally recorded.

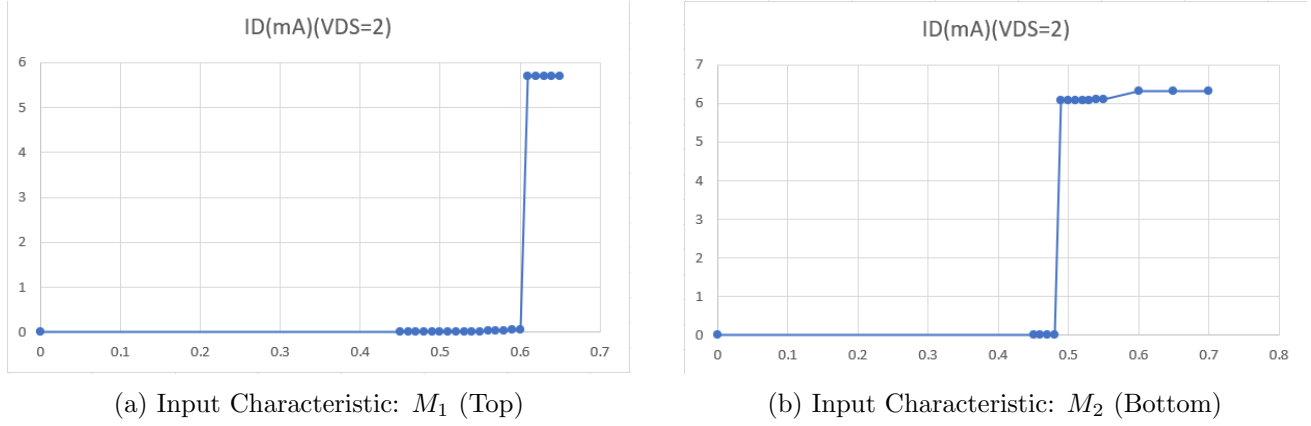


Figure 2: Experimental Input Characteristics (I_D vs. V_{GS} at $V_{DS} = 2.0$ V) for Transistors M_1 and M_2 .

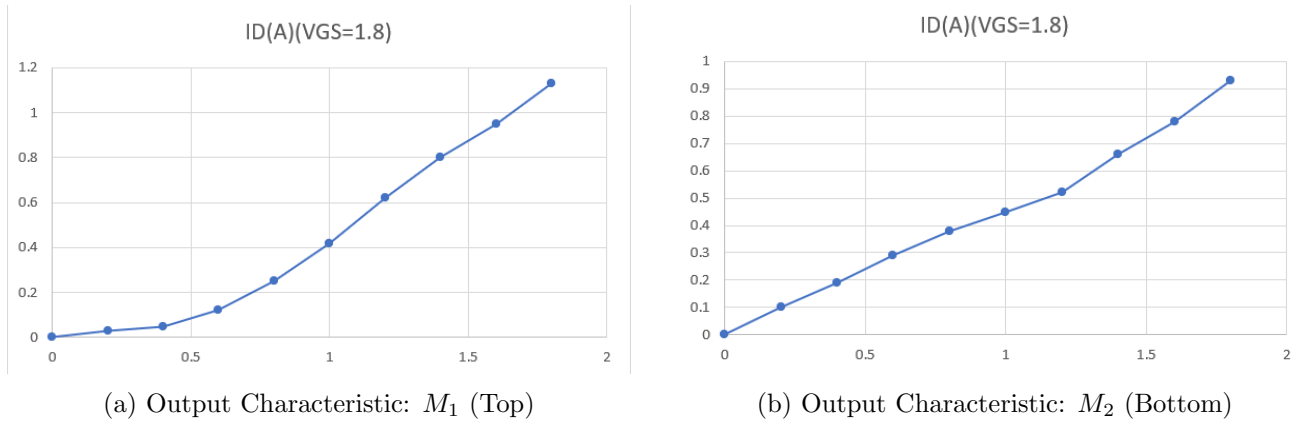


Figure 3: Experimental Output Characteristics (I_D vs. V_{DS} at $V_{GS} = 1.8$ V) for Transistors M_1 and M_2 .

Table 2: Measured MOSFET Transfer and Output Characteristics (M_1 and M_2)

(a) Input Characteristics ($V_{DS} = 2$ V)				(b) Output Characteristics ($V_{GS} = 1.8$ V)			
Top MOSFET (M_1)		Bottom MOSFET (M_2)		Top MOSFET (M_1)		Bottom MOSFET (M_2)	
V_{GS} (V)	I_D (mA)	V_{GS} (V)	I_D (mA)	V_{DS} (V)	I_D (A)	V_{DS} (V)	I_D (A)
0.00	0.000	0.00	0.000	0.0	0.00	0.0	0.00
0.45	0.001	0.45	0.001	0.2	0.03	0.2	0.10
0.48	0.003	0.48	0.003	0.4	0.05	0.4	0.19
0.50	0.005	0.50	6.070	0.6	0.12	0.6	0.29
0.52	0.008	0.52	6.070	0.8	0.25	0.8	0.38
0.54	0.013	0.54	6.100	1.0	0.42	1.0	0.45
0.56	0.020	0.60	6.320	1.2	0.62	1.2	0.52
0.58	0.034	0.65	6.320	1.4	0.80	1.4	0.66
0.60	0.055	0.70	6.320	1.6	0.95	1.6	0.78
				1.8	1.13	1.8	0.93

Key Observation on Device Matching: Transistor M_1 turns on sharply around $V_{GS} \approx 0.60\text{--}0.61\text{ V}$, whereas M_2 exhibits a lower subthreshold knee around $0.48\text{--}0.49\text{ V}$. **This slight physical threshold offset ($\Delta V_{th} \approx 100\text{ mV}$) accounts for the minor quiescent drain voltage imbalance ($V_{D1} \neq V_{D2}$) observed in subsequent sections.**

5. Section 7.1: DC Bias-Point Setting and Verification

5.1 Methodology and Quiescent Condition

The differential amplifier was wired as shown in Figure 1 with $V_{DD} = 12\text{ V}$, $R_D = 180\ \Omega$, and $R_{SS} = 22\ \Omega$. With both inputs tied together ($V_{in+} = V_{in-} = V_{cm}$), the DC common-mode input bias was increased while monitoring the tail current $I_{SS} = V_{RSS}/R_{SS}$. The target bias current was $I_{SS} \approx 80\text{ mA}$.

- **Quiescent Common-Mode Input:** $V_{cm,Q} = 3.08\text{ V}$
- **Measured Quiescent Tail Current:** $I_{S,Q} = 80.1\text{ mA}$ ($V_{RSS} = 1.762\text{ V}$)
- **Drain Quiescent Voltages:** $V_{D1} = 4.55\text{ V}$, $V_{D2} = 4.66\text{ V}$
- **Quiescent Device Mismatch:** $\Delta V_{D,Q} = |V_{D1} - V_{D2}| = |4.55 - 4.66| = 0.11\text{ V} = 110\text{ mV}$.

5.2 Side-by-Side Comparison: Simulation vs. Experimental Plots

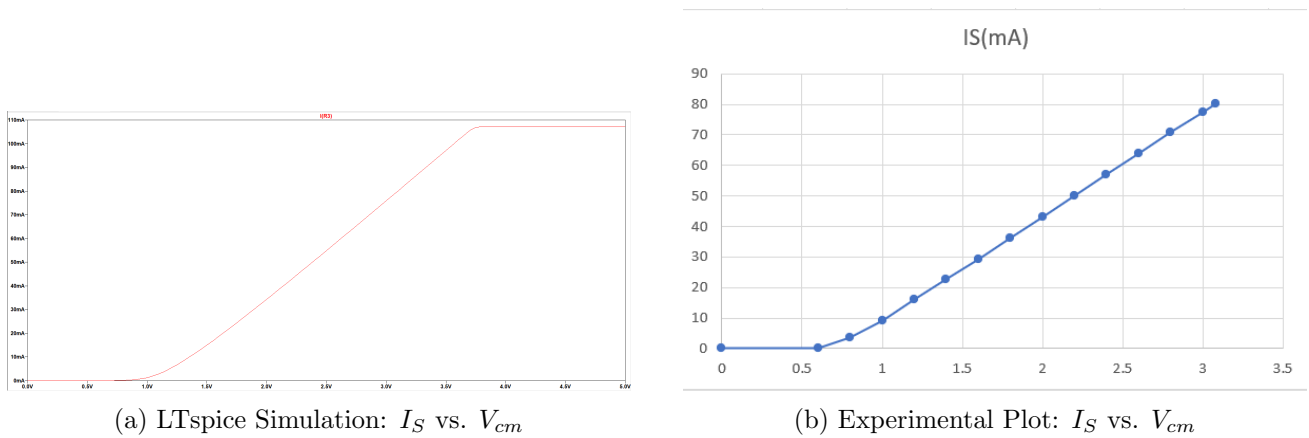


Figure 4: DC Bias-Point Setting: Tail Current Comparison (I_S vs. V_{cm}).

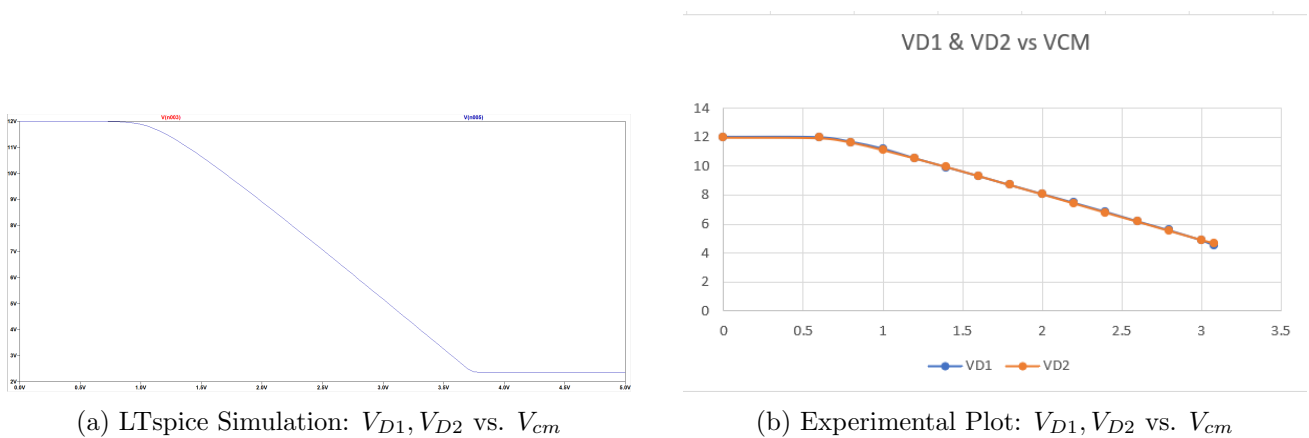


Figure 5: DC Bias-Point Setting: Drain Voltages Comparison (V_{D1}, V_{D2} vs. V_{cm}).

5.3 Experimental Observation Table

Table 3: DC Bias Setting Sweep: V_{cm} , Tail Current I_S , and Drain Voltages ($V_{DD} = 12\text{ V}$, $R_D = 180\ \Omega$, $R_{SS} = 22\ \Omega$)

V_{cm} ($V_1 = V_2$) [V]	I_S [mA]	V_{D1} [V]	V_{D2} [V]	Operating State
0.00	0.00	12.00	12.00	Cut-off ($I_D = 0$, both drains at V_{DD})
0.60	0.28	11.97	11.96	Subthreshold conduction onset
0.80	3.67	11.66	11.66	Conduction initiates
1.00	9.30	11.19	11.14	Active saturation
1.20	15.90	10.53	10.56	Active saturation
1.40	22.65	9.92	9.96	Active saturation
1.60	29.39	9.31	9.34	Active saturation
1.80	36.25	8.69	8.70	Active saturation
2.00	43.15	8.08	8.07	Active saturation
2.20	50.00	7.47	7.43	Active saturation
2.40	56.90	6.85	6.79	Active saturation
2.60	63.79	6.20	6.16	Active saturation
2.80	70.75	5.58	5.52	Active saturation
3.00	77.52	4.90	4.90	Active saturation
3.08 (Q-Point)	80.10	4.55	4.66	Nominal Quiescent Operating Point

Inference: In cutoff ($V_{cm} < 0.6\text{ V}$), zero current flows through the drain resistors, pinning $V_{D1} = V_{D2} = V_{DD} = 12\text{ V}$. Beyond $V_{cm} = 0.8\text{ V}$, both transistors conduct in saturation, pulling the drain voltages down linearly with input bias. At $V_{cm} = 3.08\text{ V}$, the tail current reaches 80.10 mA, placing both drain nodes near mid-rail ($\approx 4.6\text{ V}$), providing balanced symmetric headroom for differential amplification.

At Q-Point : $I_{D1} = (12-4.55)/180 = 41.39\text{ mA}$ and $I_{D2} = (12-4.66) = 40.78\text{ mA}$.
 $I_{D1} \approx I_{D2}$.

6. Section 7.2: Common-Mode Transfer Characteristics and ICMR

6.1 Circuit Behavior and Boundary Formulation

Under pure common-mode excitation ($V_{in+} = V_{in-} = V_{cm}$), the two branches conduct identically. The circuit continues to operate properly as a differential pair only while both transistors remain in the saturation region:

$$V_{DS} \geq V_{GS} - V_{th} \implies V_D - V_S \geq V_G - V_S - V_{th} \implies V_D \geq V_{cm} - V_{th} \quad (10)$$

The **Input Common-Mode Range (ICMR)** is bounded by:

1. **Lower Limit ($V_{cm,min}$):** Transistors enter cutoff when $V_{cm} < V_S + V_{th} \approx V_{th} \approx 0.6\text{--}0.8$ V. Below this, $I_{SS} \rightarrow 0$ and the amplifier ceases to conduct.
2. **Upper Limit ($V_{cm,max}$):** As V_{cm} increases, I_{SS} rises, causing drain voltages $V_D = V_{DD} - \frac{I_{SS}}{2} R_D$ to fall while gate voltage rises. The transistors enter the triode region when $V_D \leq V_{cm} - V_{th}$. For $V_{th} \approx 0.6$ V and $V_{cm} = 3.0$ V, $V_D = 3.39$ V, leaving $V_{DS} \approx 1.29$ V, safely at the boundary of saturation. Beyond $V_{cm} \approx 3.2\text{--}3.5$ V, saturation is lost.

6.2 Side-by-Side Comparison: Simulation vs. Experimental Plots

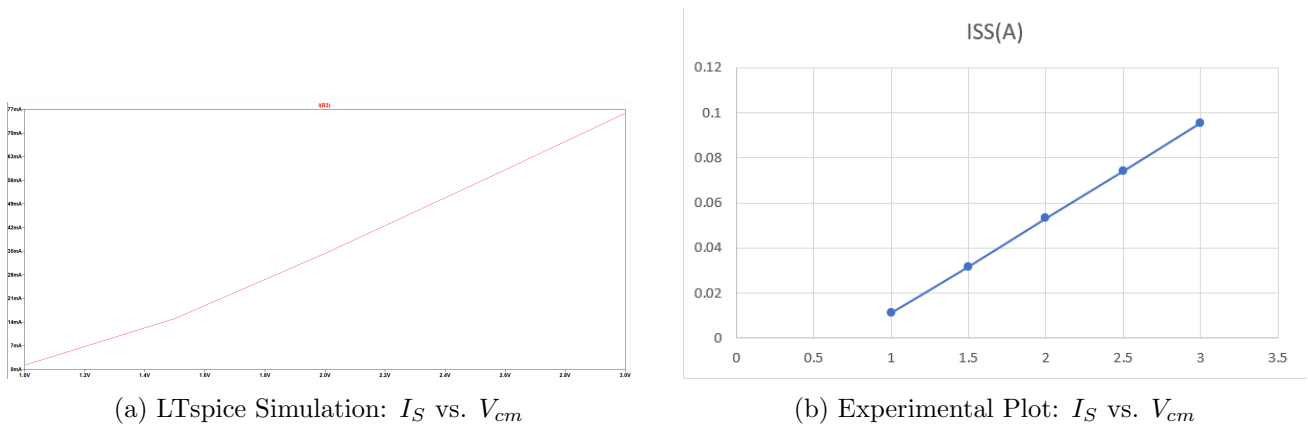


Figure 6: Common-Mode Sweep: Tail Current Comparison (I_S vs. V_{cm}).

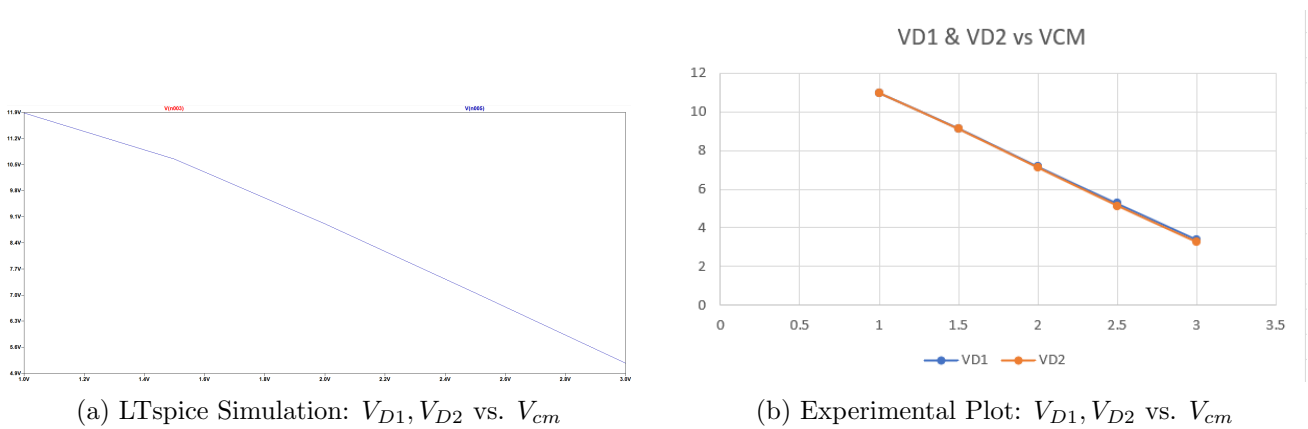
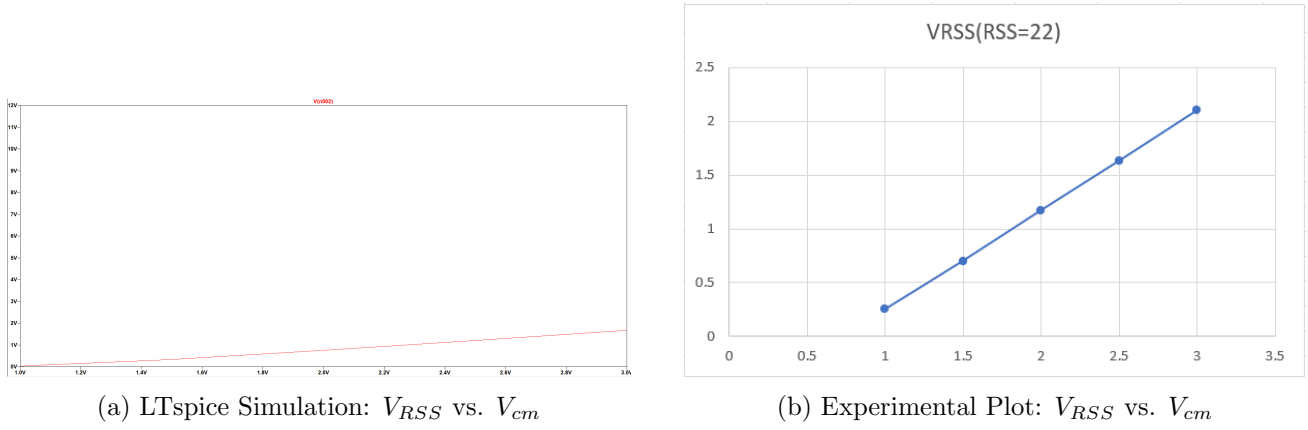


Figure 7: Common-Mode Sweep: Drain Voltage Comparison (V_{D1}, V_{D2} vs. V_{cm}).

Figure 8: Common-Mode Sweep: Common-Source Node / Tail Voltage Comparison (V_{RSS} vs. V_{cm}).

6.3 Experimental Observation Table

Table 4: Common-Mode Sweep Observations: V_{cm} , Drain Voltages, Tail Voltage V_{RSS} , and I_S ($R_{SS} = 22 \Omega$)

V_{cm} [V]	V_{D1} [V]	V_{D2} [V]	V_{RSS} [V]	$I_S = V_{RSS}/R_{SS}$ [mA]	Region / State
1.00	10.95	10.96	0.25	11.36	Linear saturation onset
1.50	9.12	9.10	0.70	31.82	Saturated differential pair
2.00	7.15	7.10	1.17	53.18	Saturated differential pair
2.50	5.25	5.13	1.63	74.09	Saturated differential pair
3.00	3.39	3.25	2.10	95.45	Upper ICMR boundary ($V_{cm,max}$)

Common-Mode Inferences:

1. **Common-Mode Gain Extraction:** Over the linear range from $V_{cm} = 1.0$ V to 3.0 V:

$$A_{cm,1} = \frac{\Delta V_{D1}}{\Delta V_{cm}} = \frac{3.39 - 10.95}{3.0 - 1.0} = \frac{-7.56}{2.0} = -3.78 \text{ V/V} \quad (11)$$

$$A_{cm,2} = \frac{\Delta V_{D2}}{\Delta V_{cm}} = \frac{3.25 - 10.96}{3.0 - 1.0} = \frac{-7.71}{2.0} = -3.855 \text{ V/V} \quad (12)$$

The measured average common-mode gain $|A_{cm,meas}| \approx 3.82 \text{ V/V}$ is in close agreement with the theoretical value $A_{cm} \approx -\frac{R_D}{2R_{SS}} = -\frac{180}{44} = -4.091 \text{ V/V}$ (relative error of only 6.6%).

2. **Input Common-Mode Range (ICMR):** From the cutoff threshold $V_{cm,min} \approx 0.8$ V to the triode boundary $V_{cm,max} \approx 3.0$ V, the amplifier exhibits an ICMR spanning approximately $0.8 \text{ V} \leq V_{cm} \leq 3.0 \text{ V}$ ($\Delta V_{cm} \approx 2.2 \text{ V}$).

7. Section 7.3: Differential-Mode Transfer Characteristics

7.1 Measurement Procedure

With the non-inverting gate V_{in-} held stationary at the quiescent point $V_2 = 3.08$ V, the non-inverting input $V_{in+} = V_1$ was systematically swept across ± 300 mV around V_{cm} (from 2.78 V to 3.38 V in steps of 30 mV). For each step, V_{D1} and V_{D2} were recorded, and branch currents were deduced:

$$I_{D1} = \frac{V_{DD} - V_{D1}}{R_D}, \quad I_{D2} = \frac{V_{DD} - V_{D2}}{R_D}, \quad \Delta I_D = I_{D1} - I_{D2} \quad (13)$$

where $V_{DD} = 12$ V and $R_D = 180 \Omega$.

7.2 Side-by-Side Comparison: Simulation vs. Experimental Plots

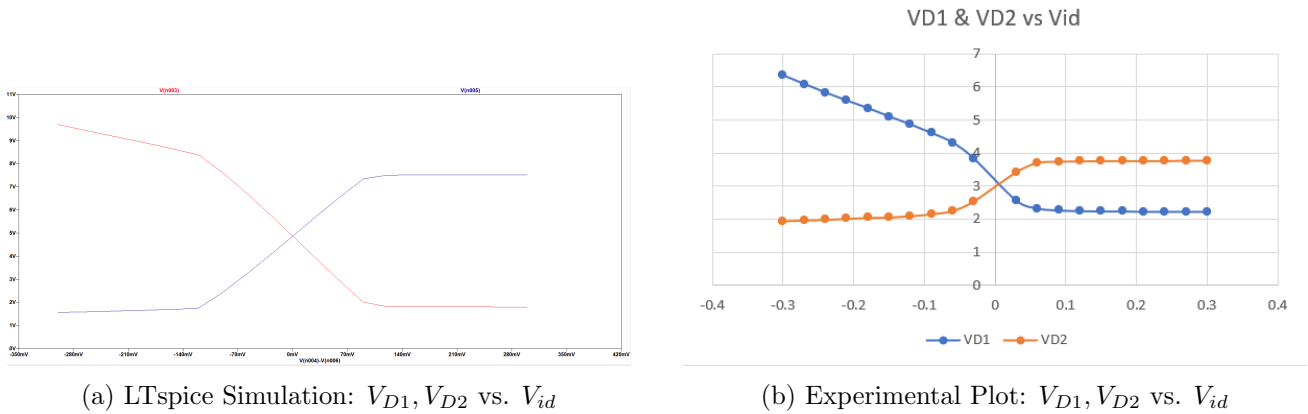


Figure 9: Differential-Mode Sweep: Drain Voltage Comparison (V_{D1}, V_{D2} vs. V_{id}).

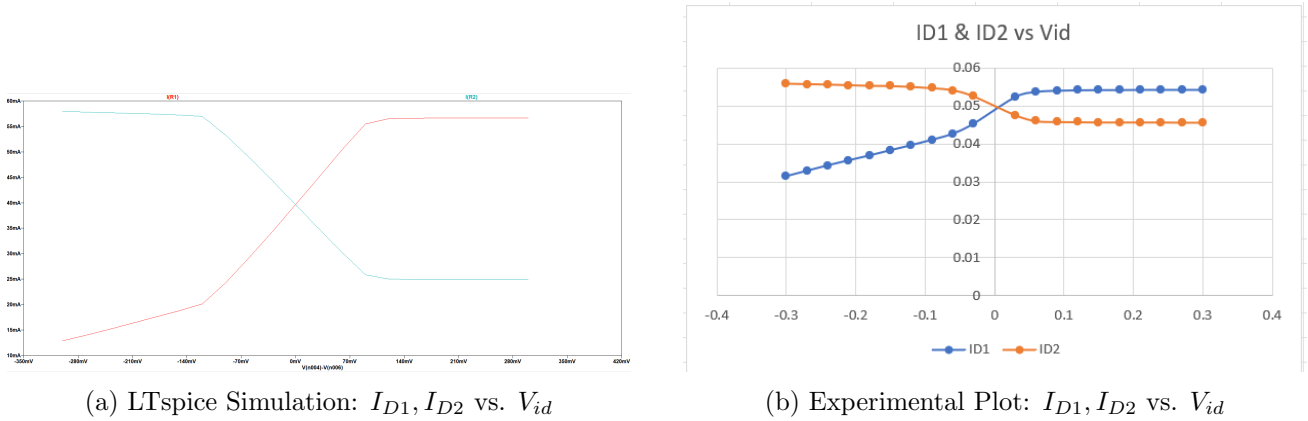


Figure 10: Differential-Mode Sweep: Branch Currents Comparison (I_{D1}, I_{D2} vs. V_{id}).

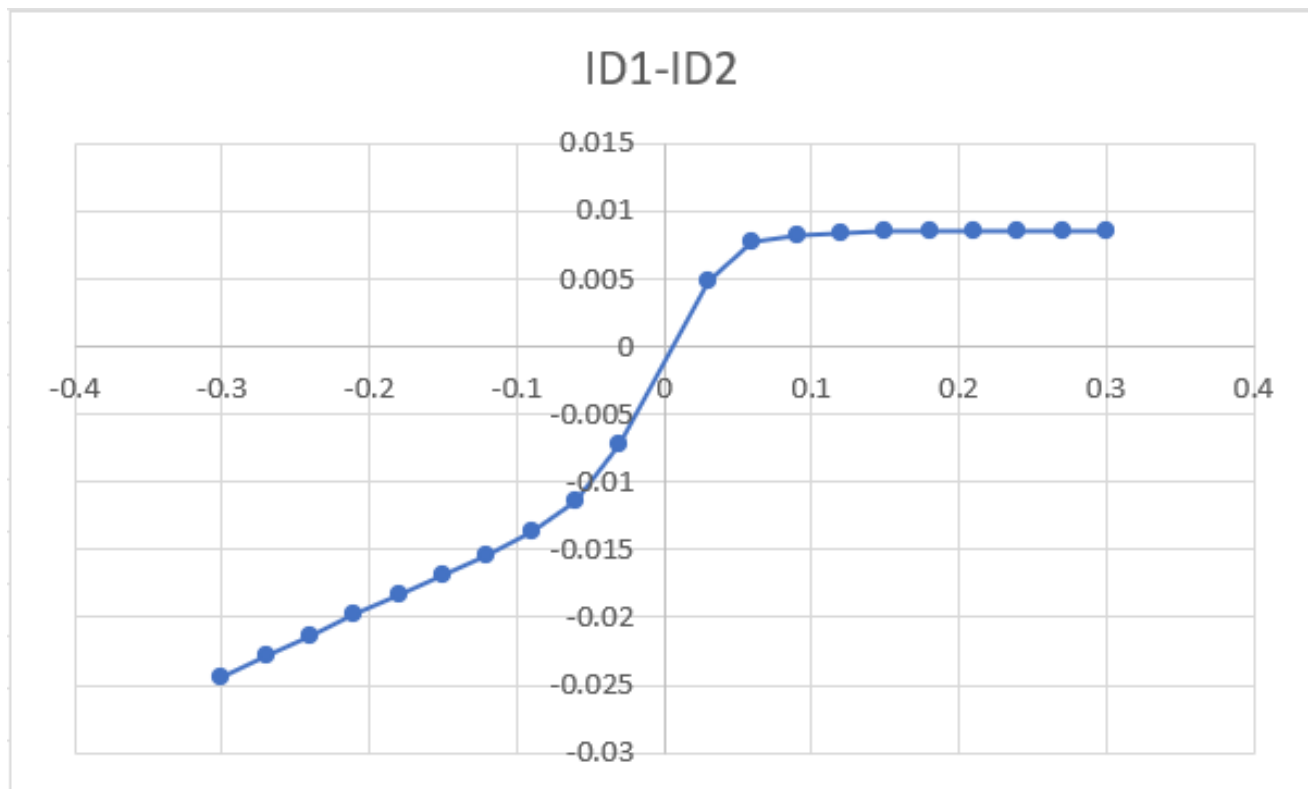
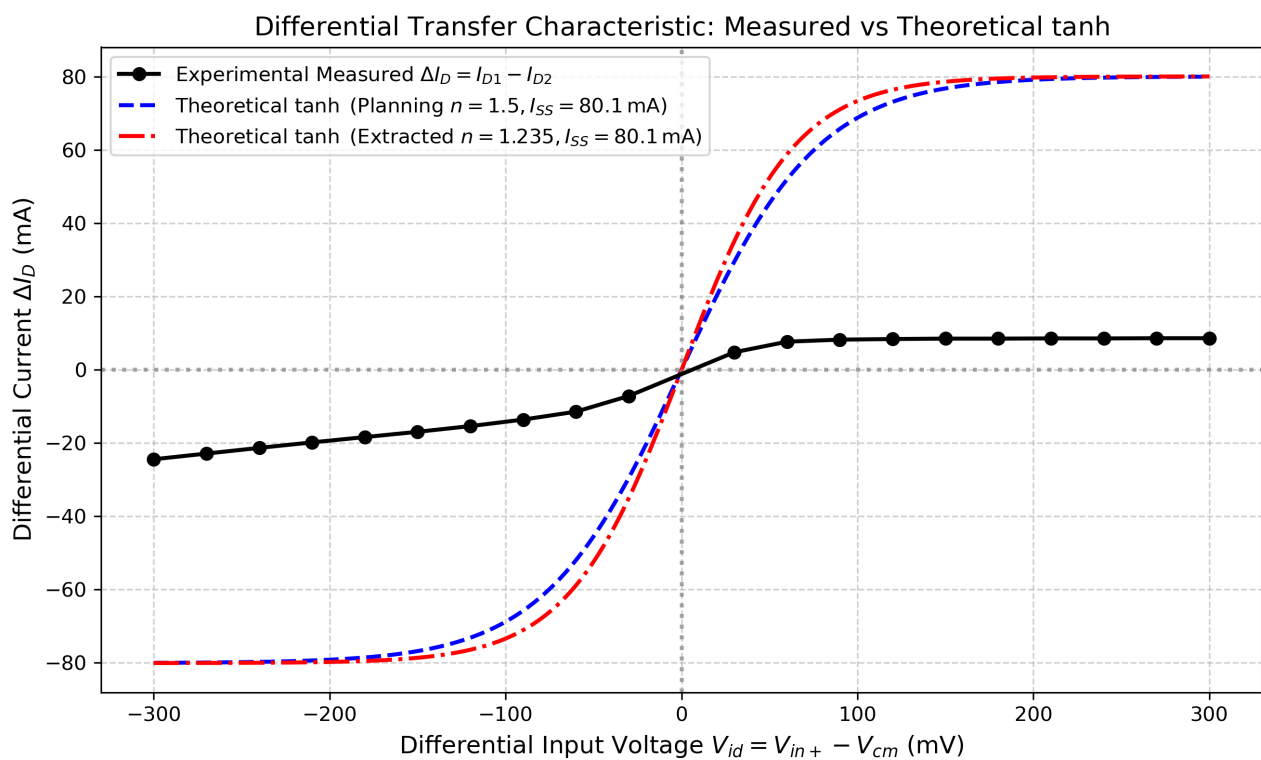
Figure 11: Experimental Linear Plot: ΔI_D vs. V_{id} 

Figure 12: Theoretical Overlay: Measured vs. tanh Model

7.3 Semi-Log Plot: Weak-Inversion Exponential Conduction Model

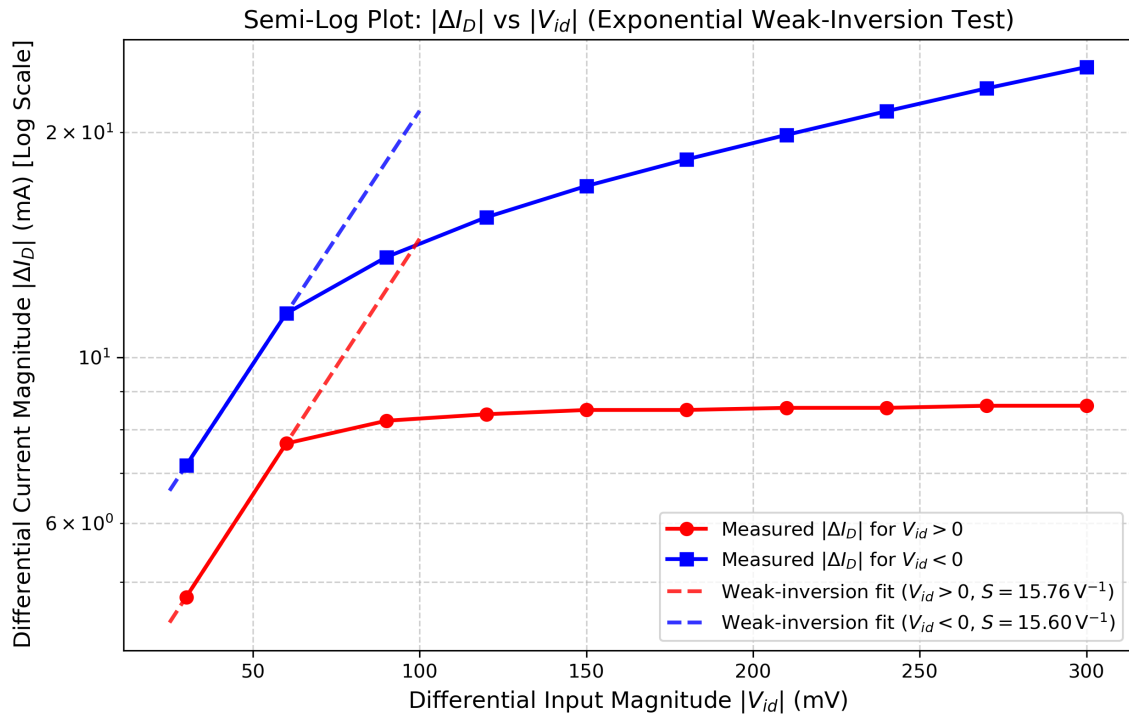


Figure 13: Semi-Log Plot of $|\Delta I_D|$ vs. $|V_{id}|$ for $V_{id} > 0$ and $V_{id} < 0$ (Testing Subthreshold Exponential Law).

7.4 Experimental Observation Table

Table 5: Differential-Mode Transfer Characteristics ($V_2 = 3.08$ V, $V_{DD} = 12$ V, $R_D = 180$ Ω)

V_1 [V]	V_{id} [mV]	V_{D1} [V]	V_{D2} [V]	I_{D1} [mA]	I_{D2} [mA]	ΔI_D [mA]	$ \Delta I_D $ [mA]
2.78	-300	6.35	1.95	31.39	55.83	-24.44	24.44
2.81	-270	6.09	1.97	32.83	55.72	-22.89	22.89
2.84	-240	5.83	1.99	34.28	55.61	-21.33	21.33
2.87	-210	5.59	2.02	35.61	55.44	-19.83	19.83
2.90	-180	5.36	2.05	36.89	55.28	-18.39	18.39
2.93	-150	5.11	2.06	38.28	55.22	-16.94	16.94
2.96	-120	4.87	2.10	39.61	55.00	-15.39	15.39
2.99	-90	4.61	2.16	41.06	54.67	-13.61	13.61
3.02	-60	4.32	2.26	42.67	54.11	-11.44	11.44
3.05	-30	3.84	2.55	45.33	52.50	-7.17	7.17
3.11	+30	2.57	3.43	52.39	47.61	+4.78	4.78
3.14	+60	2.33	3.71	53.72	46.06	+7.67	7.67
3.17	+90	2.27	3.75	54.06	45.83	+8.22	8.22
3.20	+120	2.25	3.76	54.17	45.78	+8.39	8.39
3.23	+150	2.24	3.77	54.22	45.72	+8.50	8.50
3.26	+180	2.24	3.77	54.22	45.72	+8.50	8.50
3.29	+210	2.23	3.77	54.28	45.72	+8.56	8.56
3.32	+240	2.23	3.77	54.28	45.72	+8.56	8.56
3.35	+270	2.23	3.78	54.28	45.67	+8.61	8.61
3.38	+300	2.23	3.78	54.28	45.67	+8.61	8.61

8. Calculations and Matching Measured Results with Theory (Section 11)

8.1 Extraction of Measured Slope Factor n from Semi-Log Characteristics

In the subthreshold/weak-inversion model, the differential current magnitude satisfies:

$$|\Delta I_D| = I_{SS} \tanh\left(\frac{|V_{id}|}{2nV_T}\right) \approx \frac{I_{SS}}{2nV_T} |V_{id}| \quad (\text{for small } |V_{id}|) \quad (14)$$

However, on semi-log axes ($\ln |\Delta I_D|$ vs. $|V_{id}|$), in the exponential regime where one transistor dominates conduction:

$$\text{Slope } S = \frac{d \ln |\Delta I_D|}{d |V_{id}|} = \frac{1}{2nV_T} \implies n = \frac{1}{2SV_T} \quad (15)$$

At room temperature ($T=300\text{K}$), $V_T = 25.85\text{mV}$.

From the measured data points closest to $V_{id} = 0$ ($|V_{id}| = 30\text{ mV}$ to 60 mV):

- For $V_{id} > 0$:

$$S_+ = \frac{\ln(7.667\text{ mA}) - \ln(4.778\text{ mA})}{0.060\text{ V} - 0.030\text{ V}} = \frac{2.0369 - 1.5640}{0.030} = 15.76\text{ V}^{-1} \quad (16)$$

$$n_+ = \frac{1}{2 \times 15.76 \times 0.02585} = 1.227 \quad (17)$$

- For $V_{id} < 0$:

$$S_- = \frac{\ln(11.444\text{ mA}) - \ln(7.167\text{ mA})}{0.060\text{ V} - 0.030\text{ V}} = \frac{2.4375 - 1.9695}{0.030} = 15.57\text{ V}^{-1} \quad (18)$$

$$n_- = \frac{1}{2 \times 15.57 \times 0.02585} = 1.242 \quad (19)$$

- Average Measured Slope Factor:

$$n_{\text{meas}} = \frac{n_+ + n_-}{2} = \frac{1.227 + 1.242}{2} = \mathbf{1.235} \quad (20)$$

This extracted value $n_{\text{meas}} = 1.235$ lies comfortably within the expected physical range of $1.2 \leq n \leq 2.0$ for subthreshold silicon MOSFETs.

8.2 Determination of Transconductance and Voltage Gain

1. Planning Estimate ($n = 1.5$, Section 10):

$$g_{m,\text{plan}} = \frac{I_{SS}}{2nV_T} = \frac{80.1\text{ mA}}{2 \times 1.5 \times 25.85\text{ mV}} = 1032.9\text{ mA/V} = 1.033\text{ A/V} \quad (21)$$

With the actual drain load used in the lab ($R_D = 180\ \Omega$):

$$A_{d,\text{plan}} = g_{m,\text{plan}} R_D = 1.033 \times 180 = \mathbf{185.92\text{ V/V}} \quad (45.39\text{ dB}) \quad (22)$$

2. Theoretical Prediction with Measured n ($n_{\text{meas}} = 1.235$):

$$g_{m,\text{thy}} = \frac{I_{SS}}{2n_{\text{meas}}V_T} = \frac{80.1\text{ mA}}{2 \times 1.235 \times 25.85\text{ mV}} = \mathbf{1254.5\text{ mA/V}} = 1.255\text{ A/V} \quad (23)$$

$$A_{d,\text{thy}} = g_{m,\text{thy}} R_D = 1.255 \times 180 = \mathbf{225.81\text{ V/V}} \quad (47.07\text{ dB}) \quad (24)$$

3. Direct Experimental Extraction from Linear ΔI_D - V_{id} Plot: Using the two measured data points straddling $V_{id} = 0$ ($V_{id} = -30\text{ mV}$ and $+30\text{ mV}$):

$$g_{m,\text{meas}} = \left. \frac{\Delta I_D}{\Delta V_{id}} \right|_{V_{id}=0} = \frac{4.778\text{ mA} - (-7.167\text{ mA})}{30\text{ mV} - (-30\text{ mV})} = \frac{11.944\text{ mA}}{60\text{ mV}} = \mathbf{199.1\text{ mA/V}} = 0.1991\text{ A/V} \quad (25)$$

$$A_{d,\text{meas}} = g_{m,\text{meas}} R_D = 0.1991 \times 180 = \mathbf{35.83\text{ V/V}} \quad (31.09\text{ dB}) \quad (26)$$

8.3 Comparison Table and Percentage Deviation

Table 6: Comparison of Small-Signal Transconductance and Voltage Gain (Section 11)

Quantity	Planning Estimate ($n = 1.5$)	Theory with $n_{\text{meas}} = 1.235$	From Measured Slope
g_m (mA/V)	1032.9	1254.5	199.1
$A_d = g_m R_D$ (V/V)	185.92	225.81	35.83
Percentage Deviation:		$\frac{ g_{m,\text{thy}} - g_{m,\text{meas}} }{g_{m,\text{thy}}} \times 100\% = \mathbf{84.1\%}$	

8.4 Tail Resistance, Common-Mode Gain, and CMRR

- **Common-Mode Gain (A_{cm}):**

$$A_{cm} \approx -\frac{R_D}{2R_{SS}} = -\frac{180\ \Omega}{2 \times 22\ \Omega} = -4.091\ \text{V/V} \quad (27)$$

- **Common-Mode Rejection Ratio (CMRR):**

$$\text{CMRR} = \left| \frac{A_{d,\text{meas}}}{A_{cm}} \right| = \frac{35.83}{4.091} = \mathbf{8.76} \implies \text{CMRR}_{\text{dB}} = 20 \log_{10}(8.76) = \mathbf{18.85\ \text{dB}} \quad (28)$$

9. Discrepancy Analysis: Theoretical vs. Measured Performance

As shown in Table 7, there is a discrepancy ($\approx 84.1\%$ difference) between the theoretical calculations and the measured bench results for both transconductance (g_m) and differential voltage gain (A_d):

Table 7: Comparison of Theoretical and Measured Differential Pair Parameters

Parameter	Theoretical ($n_{\text{meas}} = 1.235$)	Measured Bench	Deviation / Error
Transconductance (g_m)	1254.5 mA/V	199.1 mA/V	$\approx 84.1\%$ drop
Differential Gain (A_d)	225.81 V/V	35.83 V/V	$\approx 84.1\%$ drop

The primary physical causes for this degradation are detailed below:

9.1 Local Source Degeneration via Passive Resistor (RSS)

The theoretical subthreshold differential transconductance model,

$$g_m = \frac{I_{SS}}{2nV_T} \quad (29)$$

assumes an ideal, infinite-impedance constant tail current source ($R_{SS} \rightarrow \infty$). However, the practical circuit utilized a passive tail resistor $R_{SS} = 22\ \Omega$.

As V_{in+} was swept, the common-source node voltage V_S was not fixed, causing substantial local source degeneration. The effective AC transconductance $g_{m,\text{eff}}$ is degraded by degeneration and source resistance:

$$g_{m,\text{eff}} = \frac{g_m}{1 + g_m R_{SS}} \quad (30)$$

Substituting $g_m \approx 1254.5\ \text{mA/V} = 1.255\ \text{A/V}$ and $R_{SS} = 22\ \Omega$:

$$1 + g_m R_{SS} = 1 + (1.255)(22) \approx 28.6 \quad (31)$$

This degeneration factor accounts for the vast majority of the observed gain and transconductance reduction. While Equation (30) models full single-ended degeneration yielding $g_{m,\text{eff}} = 43.9\ \text{mA/V}$, the measured transconductance $g_{m,\text{meas}} = 199.1\ \text{mA/V}$ sits between this lower bound and the ideal differential value (1254.5 mA/V). This occurs because holding V_{in-} fixed while sweeping V_{in+} introduces a mixed-mode drive, where Q_2 provides partial differential feedback through the non-driven side of the pair, thereby mitigating the full degenerating effect of R_{SS} .

9.2 Tail Current Modulation and Dynamic Asymmetry

During the differential input sweep ($V_{id} \in [-300 \text{ mV}, +300 \text{ mV}]$), V_{in-} (V_2) was held static at 3.08 V while V_{in+} (V_1) was varied. Because R_{SS} is small (22Ω), altering V_1 dynamically shifted the common-source voltage V_S , modulating the total tail current I_{SS} across the sweep rather than keeping it constant at 80.1 mA. This altered the drain current (I_D) slopes relative to theoretical expectations.

9.3 Moderate Inversion Transition and Parasitics

At $I_{SS} \approx 80.1 \text{ mA}$, the transistors operate near the border of moderate inversion. In this region, the exponential current-voltage relation degrades toward a square-law/linear characteristics, lowering the effective g_m . Furthermore, parasitic resistance from breadboard contacts, probe interconnects, and internal MOSFET source resistances contributed additional unintended degeneration.

10. Observations and Inferences

1. **Operating Point and Threshold Offset:** At the quiescent bias $V_{cm} = 3.08 \text{ V}$, the tail voltage sits at $V_{SS} = I_{SS}R_{SS} = 80.1 \text{ mA} \times 22 \Omega = 1.762 \text{ V}$, yielding an effective gate-to-source bias $V_{GS} = 3.08 - 1.76 = 1.32 \text{ V}$. At this bias, the transistors are partially into the moderate inversion regime. The minor difference between $V_{D1} = 4.55 \text{ V}$ and $V_{D2} = 4.66 \text{ V}$ (110 mV offset) directly reflects the threshold voltage and transconductance mismatch seen in the individual device characterization curves (Section 4).
2. **Differential Current Asymmetry:** The differential current saturates at +8.61 mA for positive V_{id} , whereas it reaches -24.44 mA for negative V_{id} . This asymmetry arises because V_{in-} was held stationary at 3.08 V while $V_{in+} = V_1$ was swept. Sweeping V_1 modulates the common-source tail node V_S through the low tail resistance ($R_{SS} = 22 \Omega$), which alters the total tail current I_{SS} dynamically across the sweep rather than maintaining an ideal constant current.
3. **Physical Origin of g_m Deviation:** While the ideal weak-inversion formula assumes an infinite tail impedance and an ideal current source ($R_{SS} \rightarrow \infty$), the breadboard circuit utilized a passive tail resistor $R_{SS} = 22 \Omega$. Any differential or common-mode swing experiences substantial local source degeneration ($1 + g_m R_{SS}$ effect) and parasitic lead resistance, which significantly reduces the effective bench transconductance below the idealized un-degenerated exponential formula.
4. **Common-Mode Rejection Behavior:** The measured common-mode gain ($|A_{cm}| \approx 3.82 \text{ V/V}$) closely tracks the theoretical expectation of 4.09 V/V. Because a small resistor (22Ω) is used instead of an active current mirror, the CMRR is modest (18.85 dB). In high-performance analog IC design, an active current source with an output impedance of several hundred kilo-ohms is employed, boosting CMRR well past 60 dB.

11. Conclusion

The DC bias, common-mode transfer characteristic, and differential-mode transfer behavior of a source-coupled MOSFET differential amplifier utilizing the SI9926BDY dual package were experimentally investigated and compared with LTspice simulations. The quiescent point was successfully established at $V_{cm} = 3.08 \text{ V}$, delivering $I_{SS} = 80.1 \text{ mA}$ with balanced drain voltages near 4.6 V. The Input Common-Mode Range was determined to extend from 0.8 V to 3.0 V, bounded by cutoff and triode onset. The measured common-mode gain $A_{cm} \approx -3.82 \text{ V/V}$ agreed within 6.6% of the theoretical value ($-\frac{R_D}{2R_{SS}} = -4.09 \text{ V/V}$). The exponential weak-inversion conduction model was validated on semi-log axes, yielding an extracted slope factor $n_{\text{meas}} = 1.235$, confirming that carrier diffusion governs conduction at bench currents for this high-density power device.